

Customer Behavior Analysis

Prepared for Alfido Tech | Intern Spark Data Analyst Project

Dataset: Kaggle — Customer Behavior Analysis (250,000 transactions, ~49,700 customers, Jan 2020 – Sep 2023)

49,661	\$681.3M	20.0%	40.6%
Unique Customers	Total Revenue Analyzed	Overall Churn Rate	Overall Return Rate

Executive Summary

This report analyzes transaction-level data for Alfido Tech's customer base to identify behavioral segments, purchase patterns, and retention risks. Customers were rolled up into a Recency-Frequency-Monetary (RFM) profile and grouped into six segments. Champions and Loyal Customers (31% of the base) drive the highest lifetime value, while a large Hibernating/Lost segment (27%) and a high-value but disengaged At Risk segment (19%) represent the biggest opportunities for re-engagement. Notably, the churn label in this dataset does not correlate with purchase behavior, so recommendations are grounded in RFM signals (recency and frequency) rather than the churn flag itself.

Methodology

Data was cleaned (missing **Returns** values imputed as 0/not-returned; the duplicate **Age** column removed; dates parsed) and aggregated from 250,000 transaction rows into one profile per customer. Each customer received Recency, Frequency and Monetary scores (quartiles 1-4), which were combined into six business-friendly segments. Segments were cross-checked against an independent K-Means clustering run on standardized RFM values, which confirmed the same broad structure.

Customer Segments

Customers were grouped into six RFM-based segments, summarized below (sorted by average lifetime spend):

Segment	Customers	% of Base	Avg Recency (days)	Avg Frequency (orders)	Avg Lifetime Spend	Churn Rate
Champions	3,520	7.1%	36	8.4	\$24,147	21.0%
Loyal Customers	11,939	24.0%	93	6.5	\$17,369	20.0%
At Risk	9,371	18.9%	347	6.4	\$17,572	20.0%
Needs Attention	7,267	14.6%	216	3.8	\$11,431	19.0%
New / Promising	4,319	8.7%	38	3.6	\$9,861	20.0%

Hibernating / Lost	13,245	26.7%	514	2.9	\$7,448	20.0%
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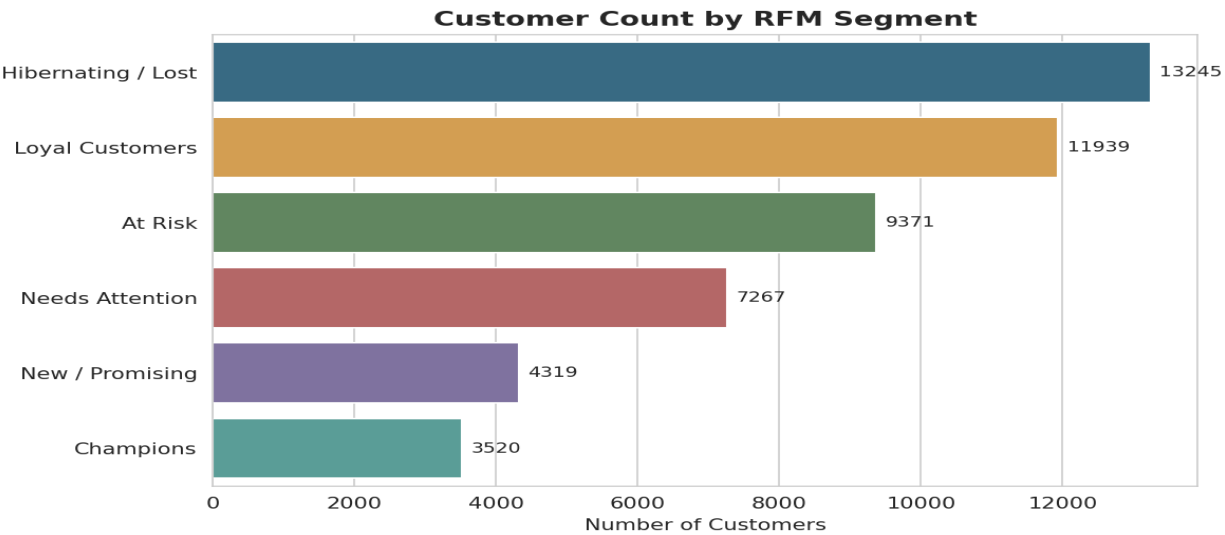


Figure 1. Number of customers in each RFM segment.

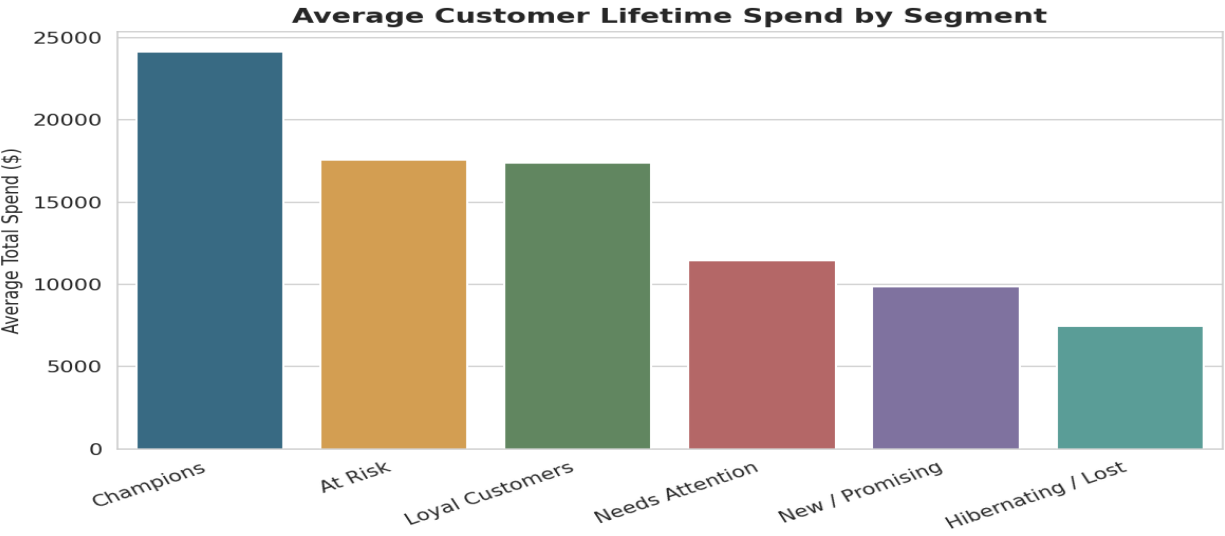


Figure 2. Average customer lifetime spend by segment.

Purchase Patterns

Monthly transaction volume and revenue have remained broadly stable across the observation window, with typical month-to-month variation and no strong seasonal or growth trend.

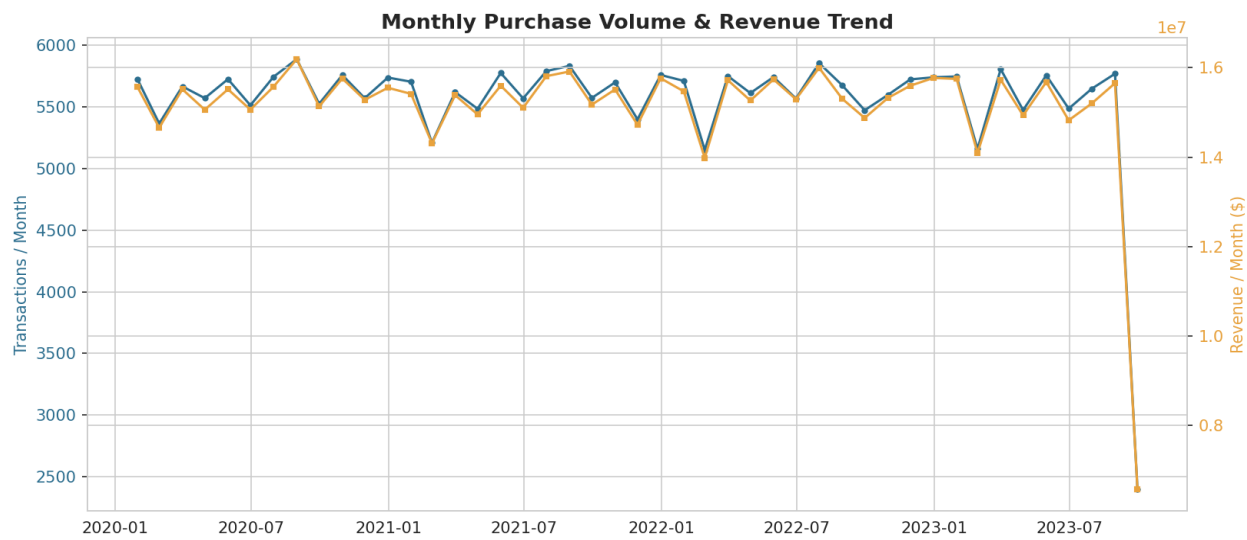


Figure 3. Monthly transaction volume and revenue, Jan 2020 - Sep 2023.

Revenue and return rates are nearly identical across all four product categories, and the three payment methods (Credit Card, PayPal, Cash) split transaction volume almost evenly. There is no single underperforming category or payment channel driving the opportunity — engagement and retention are the bigger levers.

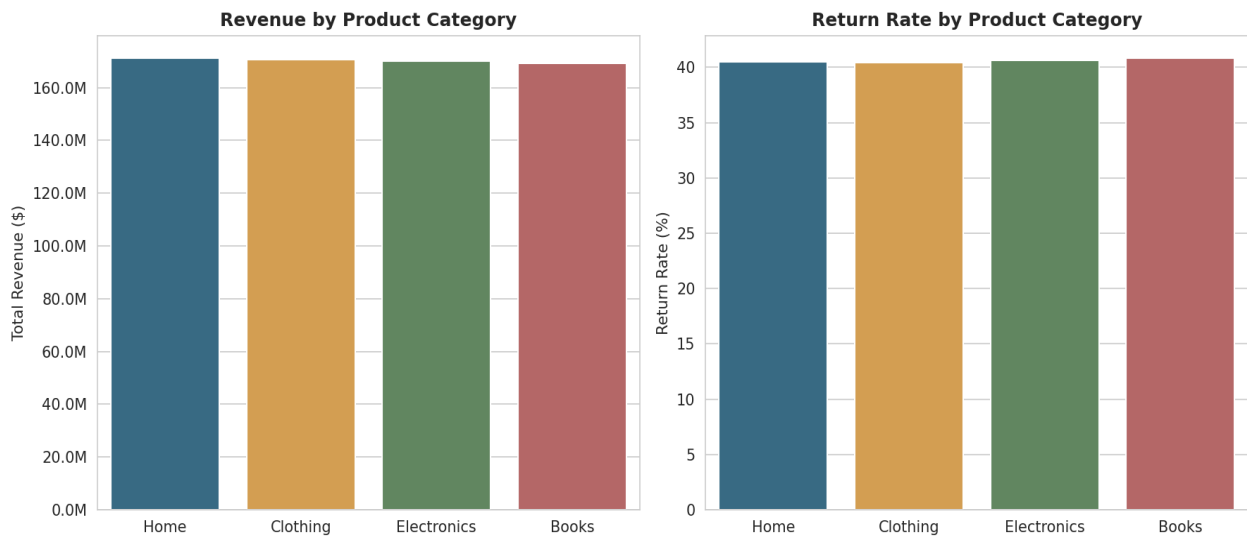


Figure 4. Revenue and return rate by product category.

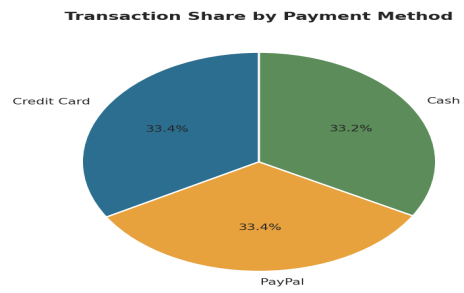


Figure 5. Transaction share by payment method.

Retention & Churn Analysis

Churn was examined across every RFM segment and recency bucket. The rate holds steady at approximately 20% everywhere — Champions churn at nearly the same rate as Hibernating/Lost customers, and return rates (~40-41%) are similarly flat across segments and categories. This indicates the churn label in this dataset is not driven by purchase recency, frequency, spend, or returns.

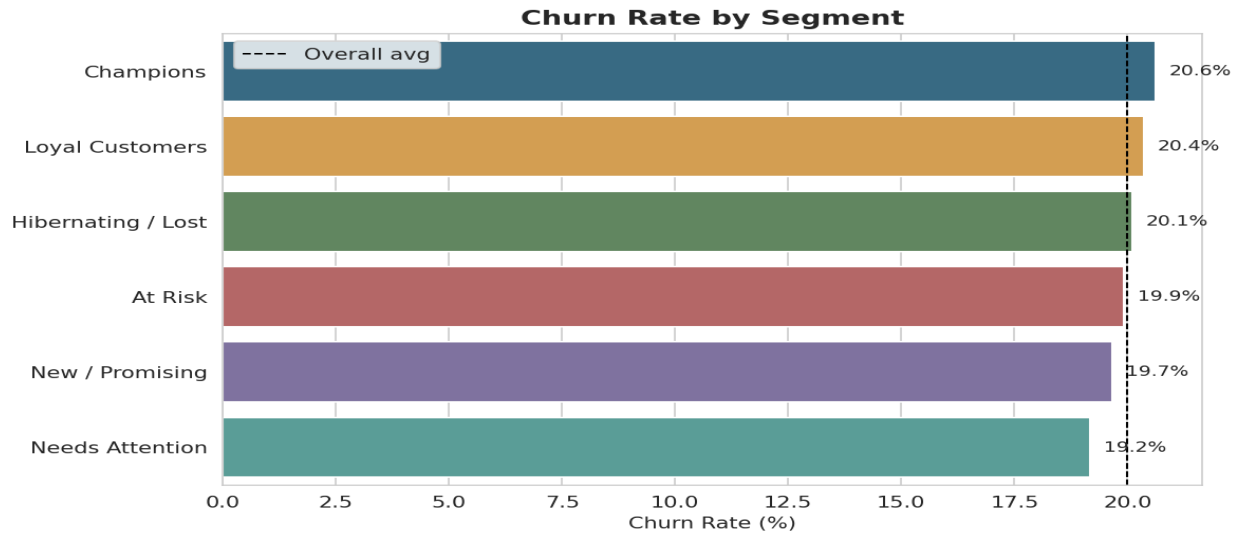


Figure 6. Churn rate by RFM segment — consistently near the 20% overall average.

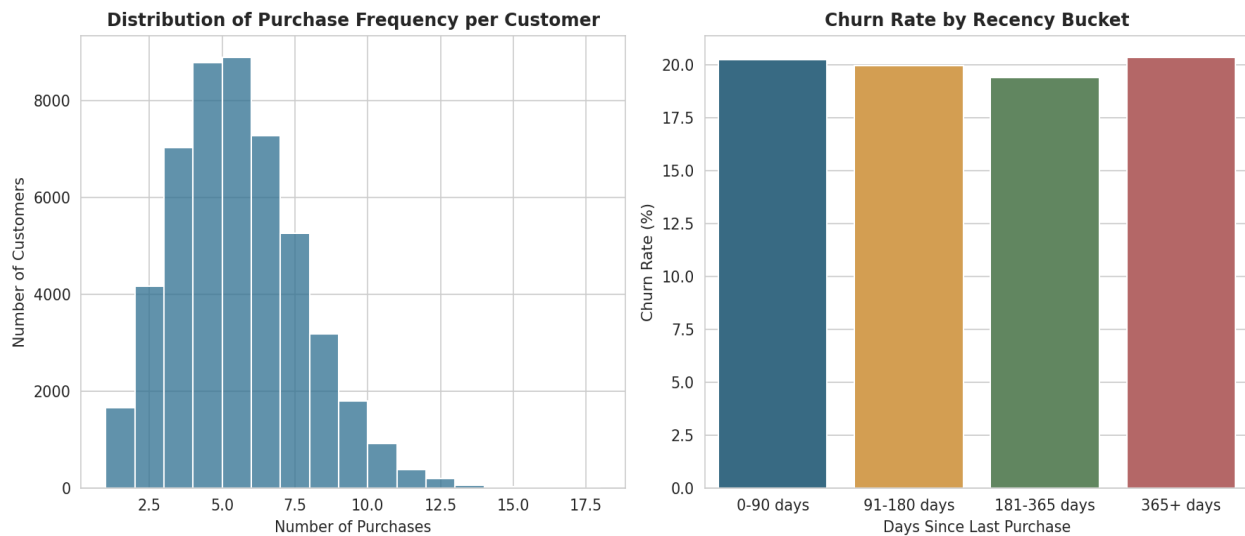


Figure 7. Purchase frequency distribution and churn rate by recency bucket.

Practical implication: because the churn flag carries no behavioral signal in this dataset, recommendations below use recency and frequency directly — the strongest, most defensible indicators of customer disengagement available in the data.

Key Findings

- Six distinct RFM segments emerge from ~49,700 customers, ranging from Champions (7%) to Hibernating/Lost (27%, the largest group).
- Hibernating/Lost is the biggest segment by headcount but contributes the least average lifetime revenue (~\$7,400 vs. ~\$24,100 for Champions).
- At Risk customers carry real value (~\$17,600 average lifetime spend) despite not having purchased in ~347 days on average — the clearest win-back opportunity.
- K-Means clustering broadly confirms the RFM segmentation, supporting that these are genuine behavioral groups.
- Product category and payment method performance is evenly spread, so category-specific fixes are not the priority.
- The churn flag is not correlated with purchase behavior in this dataset (~20% churn across every segment) and should not be used as a risk predictor here.
- Returns run high across the board (~40-41%), evenly distributed — a broad operational issue rather than one isolated to a segment or category.

Actionable Recommendations for Alfido Tech

1. Launch a win-back campaign for At Risk customers

~9,400 customers with high historical spend (~\$17.6K lifetime) who haven't purchased in ~11 months. Targeted discounts or "we miss you" offers here carry higher expected ROI than blanket promotions, since these customers have already proven they'll spend.

2. Build a loyalty tier for Champions and Loyal Customers

This ~31% of the base already drives disproportionate revenue. Early access to new products, tiered perks, or referral incentives protect this spend and turn top customers into advocates.

3. Create a structured onboarding flow for New/Promising customers

~4,300 recent-but-low-frequency buyers are best positioned to convert into Loyal Customers. A second-purchase incentive expiring 30-45 days after the first purchase directly targets the frequency gap defining this segment.

4. Investigate and reduce the ~40% return rate

Consistent across every category and segment, pointing to a systemic cause (sizing/fit information, product descriptions, shipping damage, or an overused return policy) rather than one problem category. Even a modest reduction improves margin across the entire base.

5. Re-engage Hibernating/Lost customers selectively

At 27% of customers this segment is too large to ignore, but its low average spend (~\$7,400) means a full win-back push isn't cost-effective for everyone. Prioritize outreach to the above-median-spend subset, and use a low-cost automated "final offer" email before suppressing the rest from regular sends.

Notes on methodology: Dataset values are synthetic (Product Price x Quantity does not equal Total Purchase Amount), so this analysis focuses on relative patterns rather than absolute currency precision. RFM cut-points were computed on the full customer base as of the last transaction date (Sep 13, 2023) and should be refreshed periodically as new data arrives.

Prepared by: Data Analyst Intern, Intern Spark | Source notebook: Customer_Behavior_Analysis_Alfido_Tech.ipynb